

PROFESSIONAL SUMMARY:

AI engineer specializing in multimodal foundation models, Vision-Language-Action (VLA) systems, robotic manipulation, and embodied AI. Hands-on experience developing end-to-end AI systems, fine-tuning foundation models, building scalable multimodal training pipelines, and optimizing distributed GPU workloads on NVIDIA H100 platforms. Experienced with PyTorch, CUDA, computer vision, imitation learning, and robot learning, from simulation to real-world deployment. Author of two peer-reviewed publications in machine learning and artificial intelligence.

EDUCATION:

Masters of Science in Artificial Intelligence, 2026

San Jose State University, San Jose, CA, USA

Master of Applied Science (MAsc) - Electrical and Computer Engineering, 2023

Toronto Metropolitan University (Formerly Ryerson University), Toronto, ON, Canada

Bachelor of Engineering Degree - Computer Engineering, 2022

Toronto Metropolitan University (Formerly Ryerson University), Toronto, ON, Canada

HIGHLIGHT OF QUALIFICATIONS AND SKILLS:

Machine Learning: Deep Learning, Transformer Models, Large Language Models (LLMs), Vision-Language Models (VLMs), Vision-Language-Action (VLA), Computer Vision, Imitation Learning, Reinforcement Learning, Multimodal Learning, Robotic Manipulation

Frameworks: PyTorch, TensorFlow, LeRobot, Hugging Face, LangChain, OpenAI API

Robotics: NVIDIA GR00T N1.5, Isaac Sim, LIBERO, ROS 2, Robot Learning, Data Collection, Simulation

Programming: Python, C/C++, CUDA, Java, Shell Scripting

Infrastructure: Linux, Git, AWS, GPU Programming, Distributed Training, NVIDIA H100, NVIDIA RTX GPUs

PROFESSIONAL EXPERIENCE:

AI Research Assistant, San Jose State University, San Jose, CA, USA, 2025 - 2026

- Fine-tuned NVIDIA GR00T N1.5 Vision-Language-Action foundation models for robotic manipulation using PyTorch and LeRobot on NVIDIA H100 (96 GB) GPUs.
- Built scalable multimodal training pipelines supporting 20+ robot demonstration episodes from simulation and real-world robotic manipulation datasets.
- Developed a Vision-Language-Model (VLM)-based instruction augmentation pipeline that automatically converts robot observations into structured task descriptions, improving policy learning.
- Evaluated robotic policies on LIBERO benchmarks, Isaac Sim, and physical robots using success rate, execution time, MAE, and MSE metrics.
- Improved robotic manipulation success rate from 26.7% (short-command fine-tuning) to 56.7% using prompt-engineered instruction augmentation (2.1× improvement).
- Optimized large-scale fine-tuning workflows on H100 GPUs through mixed-precision training, efficient data loading, and GPU memory optimization.

AI Research Assistant, Toronto Metropolitan University, Toronto, SIP Lab, ON, 2021 – 2024

- Designed and trained deep learning models for regression and predictive modeling using PyTorch and TensorFlow.

- Conducted neural network architecture analysis and hyperparameter optimization to improve model performance and generalization.
- Developed Python-based machine learning pipelines for data preprocessing, model training, and evaluation.
- Published peer-reviewed research in machine learning and statistical optimization.

SoC Verification Engineer, Co-op, AMD, Markham, ON, 2020 – 2021

- Developed automated verification tools for CPU and GPU hardware validation using C.
- Designed testing frameworks to improve verification efficiency and test coverage.
- Applied knowledge of computer architecture to verify complex processor designs.
- Collaborated with cross-functional hardware and software engineering teams to debug and resolve verification issues.

PROFESSIONAL PROJECT EXPERIENCE:

Vision-Language-Action (VLA) Robotic Manipulation

Python • PyTorch • NVIDIA GR00T N1.5 • LeRobot • CUDA • Isaac Sim • ROS 2 • LIBERO

- Fine-tuned NVIDIA GR00T N1.5 foundation models for robotic manipulation using imitation learning and multimodal datasets.
- Built robot learning datasets containing synchronized robot states, actions, and multi-camera observations from simulation and physical robots.
- Developed an automated Vision-Language-Model pipeline that generates structured task instructions for Vision-Language-Action policy learning.
- Increased robotic task success rate from 26.7% to 56.7% through multimodal instruction augmentation and model fine-tuning.
- Evaluated models across 30 real-world robot trials per model using task success rate, execution time, MAE, and MSE.
- Trained and optimized large multimodal models on NVIDIA H100 GPUs, improving training efficiency through optimized data pipelines and mixed-precision training.

Self-Driving Car Object Detection

Python • PyTorch • YOLO • KITTI • Tesla T4

- Developed object detection models for autonomous driving using the KITTI dataset.
- Improved inference speed and detection accuracy by transitioning from Faster R-CNN to YOLO.
- Evaluated detection performance using COCO metrics and optimized training on NVIDIA Tesla T4 GPUs.

RAG-Based AI Coder

Python • LangChain • OpenAI API

- Developed a Retrieval-Augmented Generation (RAG) system for AI-assisted software development.
- Integrated LangChain with OpenAI APIs to improve code generation and contextual reasoning.
- Built document indexing and retrieval pipelines to improve response accuracy and retrieval efficiency

PUBLICATION:

Neural Network Regression Structural Hyperparameter Selection Using Reconstruction Error Minimization (REM) - *MDPI 2026*

Polynomial Regression Hyperparameter Selection and Analysis using Reconstruction Error Minimization (REM) - *IEEE Swiss Conference on Data Science 2024*